# Quantum Black Holes and The Gravitational Path Integral

**Paul Haskins**, supervised by Prof. **Jan Manschot**.

## Why Quantum Black Holes?

In contemporary physics there is a significant effort to reconcile quantum mechanics with general relativity to form a theory of quantum gravity. Black Holes provide a natural laboratory to test such theories.
Classically the entropy of a black hole $\mathcal{S}_{BH}$ was postulated by Bekenstein to be proportional to the horizon area $A$ [1]

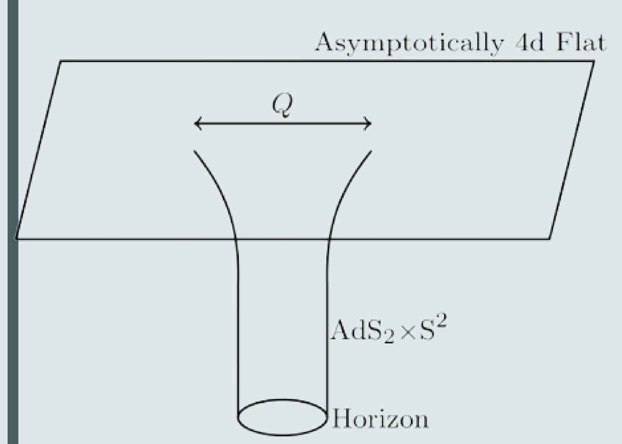
$$\mathcal{S}_{BH} = \frac{Ak_B}{4\ell_p^2} \qquad \ell_p := \sqrt{\frac{\hbar G}{c^3}}$$

This led physicists on a decades long quest to develop a quantum description of the entropy.
It was Strominger and Vafa who in 1996 discovered a microscopic description for the case of 5 d black holes in a $D1 - D5$ brane system[2].
This is the first instance of the theme of **matching a microscopic and a macroscopic description** which is the prevailing idea throughout the field.

## Reissner–Nordström (RN) Black Holes

RN black holes carry an electric and magnetic charge $q$ and $p$. The metric of such a black hole is

$$ds^2 = -\left(1 - \frac{2M}{r} + \frac{p^2+q^2}{r^2}\right) dt^2 + \left(1 - \frac{2M}{r} + \frac{p^2+q^2}{r^2}\right)^{-1} dr^2 + r^2 d\Omega_2$$




The extremal limit coincides with the so-called BPS bound

$$M^2 = p^2 + q^2$$

We study these BPS-protected black holes in the context of four-dimensional $\mathcal{N} = 8$ supergravity. The near-horizon region of these black holes is $AdS_2 \times S^2$.
The Bekenstein–Hawking entropy of these black holes in $1/8\,\mathcal{N} = 8$ string theory was found to take the form [3]

$$S_{BH}(\Delta) = \pi\sqrt{\Delta(n,\ell)}, \qquad \Delta := 4n - \ell^2 \qquad n \in \mathbb{N},\ \ell \in \mathbb{Z} \tag{1}$$

This equivalent formulation of the entropy is what we shall try to replicate from both a microscopic and macroscopic point of view.

## $\mathcal{N} = 8$ Microscopic Interpretation

We can define the variables

$$q := e^{2\pi i\tau} \quad \zeta := e^{2\pi i z} \qquad z \in \mathbb{C},\ \tau \in \mathbb{H}, \quad \mathbb{H} := \{\tau \in \mathbb{C} |\ \mathrm{Im}(\tau) > 0\}$$

The generating function of the degeneracy of BPS states is given by [3]

$$\varphi(\tau,z) = \frac{\vartheta_1(\tau,z)^2}{\eta^6(\tau)} = \sum_{n,\ell\in\mathbb{Z}} c(n,\ell)\, q^n \zeta^\ell.$$

where we define the Jacobi theta function of weight one $\vartheta_1(\tau,z)$ and the Dedekind eta function $\eta(\tau)$ defined as

$$\vartheta_1(\tau,z) := \sum_{n\in\mathbb{Z}+\frac{1}{2}} (-1)^n q^{\frac{n^2}{2}} \zeta^n, \qquad \eta(\tau) := q^{\frac{1}{24}} \prod_{n=1}^{\infty} (1-q^n).$$

We can decompose $\varphi$ into theta functions, obtain a vector-valued modular form $h_\nu$ which can be used in a generalisation of Hardy-Ramanujan-Rademacher expansion

$$C(\Delta) = \frac{1}{2\pi i} \oint_{|q|=r} \frac{h_\nu(\tau)}{q^{\frac{\Delta}{4}+1}} dq.$$

Integrating over the Ford circles centered on the Farey fractions leads to

$$C(\Delta) = 2\pi \left(\frac{\pi}{2}\right)^{\frac{7}{2}} \sum_{c=1}^{\infty} c^{-\frac{9}{2}} K_c(\Delta)\, \tilde{\mathcal{I}}_{\frac{7}{2}}\left(\frac{\pi\sqrt{\Delta}}{c}\right). \tag{2}$$

Accounting for the Goldstino zero-modes from SUSY breaking we obtain

$$W_{\text{micro}}(\Delta) = (-1)^{\Delta+1} C(\Delta).$$

which in the logarithm of the asymptotic ($\Delta \to \infty$) expansion gives us (1).

## $\mathcal{N} = 8$ Macroscopic Interpretation

**The gravitational path integral** is the adaptation of Feynman's path integral formalism to gravity, developed in the context of black hole thermodynamics by Gibbons and Hawking. One needs to define boundary conditions not just for the fields but also for the geometry. It is an inherently infinite dimensional integral. As such, techniques such as **equivariant localisation** are implemented to make it tractable.

$$\mathcal{W}(q,p) = \int [\mathcal{D}g\mathcal{D}\Psi\mathcal{D}A\mathcal{D}\Phi]\, e^{-S^{\text{bulk}}_{\text{sugra}} - S^{\text{bdry}}_{\text{sugra}}}$$

where one integrates over the metric $\mathcal{D}g$, the gravitini $\mathcal{D}\Psi$, the gauge fields $\mathcal{D}A$ and the matter fields $\mathcal{D}\Phi$ in the theory.
Localising on $AdS_2 \times S^2$ we obtain the **quantum entropy function**

$$\mathcal{W}^c = \int \left[d^{n_V+1}\phi\right]_c \mathcal{Z}^{Q_{\text{eq}}\mathcal{V}}_{\text{1-loop}} \mathcal{Z}^{\text{top}}_c \exp\left[-\frac{\pi q_\Lambda \phi^\Lambda}{c} + \frac{4\pi}{c}\,\mathrm{Im}\,F\left(\frac{\phi^\Lambda + ip^\Lambda}{2}\right)\right]$$

where $\Lambda = 0, \ldots, n_V$, $F$ is the prepotential and $n_V$ is the number of vector multiplets.
For $\mathcal{N} = 8$ supergravity the field content is given by one graviton multiplet, six gravitino multiplets, 15 vector multiplets and ten hypermultiplets, which gives us 28 gauge fields and 70 scalars.

$$W^c(q,p) = \int d^{16}\phi \sqrt{\frac{-2C_{IJK}p^Ip^Jp^K\ \det(3C_{IJ})}{c^{16}(\phi^0)^{18}}}\, \frac{1}{c}\left(\frac{\phi^0}{-2C_{IJK}p^Ip^Jp^K}\right)^4 \sqrt{c}K_c(\Delta)$$
$$\cdot \exp\left[-\frac{\pi C_{IJK}p^Ip^Jp^K}{c\phi^0} + \frac{3\pi C_{IJK}\phi^I\phi^Jp^K}{c\phi^0} - \frac{\pi q_0\phi^0}{c} - \frac{\pi q_I\phi^I}{c}\right]$$

Separating zero modes by introducing the indices $\{I,J,K = 1,\ldots,n_V\}$ we take the same limit and retrieve (1).

## A Reduction of Supersymmetry

Having understood the $\mathcal{N} = 8$ case, the natural question is what happens if we reduce the supersymmetry to $\mathcal{N} = 4$ or $\mathcal{N} = 2$?
For the microscopic side of the former case, the result was obtained in terms of mock modular forms, multiplier systems, Kloosterman sums and Eichler integrals [4]. However, a macroscopic picture is yet to be formulated. There are complications due to the emergence of wall-crossing and so called multi-centred contributions.
For the latter case, there are even more details to pin down. One example is the fact that the precise form holomorphic prepotential is not known to all orders.

## Conclusions

- The matching of the string theory microstate counting with the Bekenstein-Hawking entropy formula has been a significant milestone in our understanding of black hole thermodynamics.
- The developments of [5] have been verified, which relied on deep mathematical structures such as CFTs, automorphic forms and the Hardy-Ramanujan Rademacher expansion.
- There are several areas which warrant further investigation such as formulating a description in less supersymmetric settings. Additionally, the extension of these results to non-supersymmetric black holes remains an open question.

## References

Trinity College Dublin
The University of Dublin